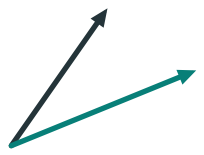
01

Bring them together.

They match.

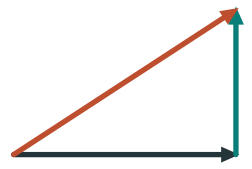


Now there is a difference.

02

Take this step.

Then take this step.



One step reaches the same end.

03

A few steps.

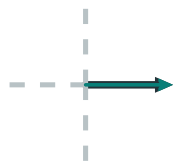
More, smaller steps.

Follow the change more closely.

04

Start together.

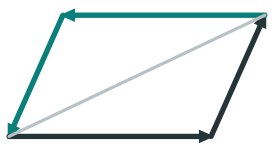
Turn just one.



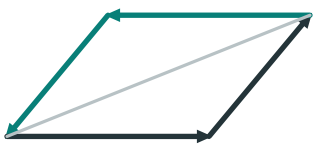
A full turn brings it back.

05

Two pieces make this patch.

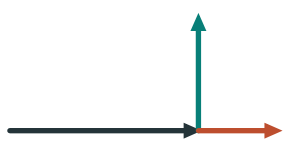


Slide the top sideways.



Same area. Different shape.

06

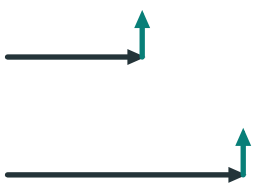


Green sets black’s next change.

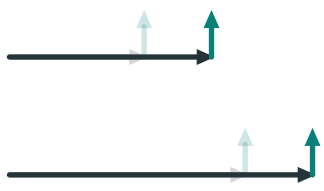
Black sets green’s opposite change.

Together, the pair turns.

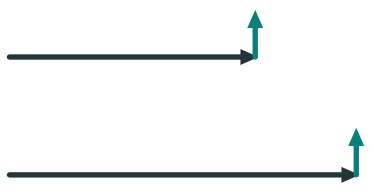
07



New rule: slide at green’s pace.



Start farther along.



The same shift survives the move.

08

Black advances.

Green stays the same.

Every step keeps that amount.

The mathematics and the debt

Companion to strips 01–08 · supplied structure, derived consequences, explicit limits

What these pictures can establish

This is a supplied classical model with one degree of freedom. It makes a local Hamiltonian construction and one Noether example visible. It does not derive mechanics from marks alone, nor yet explain quantum mechanics.

01–04 · Comparing, composing, refining, returning

We supply an oriented affine plane and comparison by translation. Tip-to-tail composition is vector addition in that plane. Refinement samples a prescribed differentiable path; finite drawings do not prove a limit or smoothness. A supplied reversible rotation gives cyclic relative position. That is geometric phase, not a quantum amplitude. Colours distinguish marks; they do not specify a physical ontology.

05 · The extra structure that makes Hamilton possible

For the remaining strips we choose a state plane with dimensionless conjugate coordinates q,p and oriented area $\omega = dq \wedge dp$. This pairing is an assumption: perpendicular ink strokes do not establish it. With physical units, $p\,dq$ has units of action; momentum need not equal velocity. The patch is the object being transformed, not a background container. Its orientation matters.

05 → 06 · The local derivation

Assume an autonomous C^1 field $X=(u,v)$, its differentiable local flow, and preservation of ω for EVERY sufficiently small cell. One sheared patch only illustrates this hypothesis. Area preservation gives $u_q + v_p = 0$. Hence $\alpha = \iota_X\omega = u\,dp - v\,dq$ is closed: $d\alpha = 0$. On a simply connected local patch $\alpha = dH$, so

$$\dot{q} = u = \partial H/\partial p, \qquad \dot{p} = v = -\partial H/\partial q.$$

The Hamiltonian is reconstructed up to an additive constant. This is local: topology may obstruct a globally single-valued H. Calling H physical energy requires a physical model.

06 · An exact turning rule

Choose $H=(q^2+p^2)/2$. Then $\dot{q}=p$ and $\dot{p}=-q$. The orange arrows depict instantaneous changes, not consecutive finite kicks. For a time h the exact update is

$$q' = q\cos h + p\sin h, \qquad p' = p\cos h - q\sin h.$$

It preserves both oriented area and H. The naive Euler update $(q+h\,p,\ p-h\,q)$ instead multiplies area and q^2+p^2 by $1+h^2$. Cyclic motion belongs to this example, not to every Hamiltonian flow.

07 · A different rule and its continuous symmetry

Now choose $H=p^2/2$. The exact evolution is $\Phi_t(q,p)=(q+t\,p,p)$. Shifting the starting configuration by $T_a(q,p)=(q+a,p)$ commutes with that evolution: $\Phi_t\,T_a=T_a\,\Phi_t$. This canonical translation leaves H unchanged. Strip 7 switches rules; the oscillator of strip 6 does NOT have this translation symmetry.

08 · The Noether charge and why it is conserved

The translation generator $Y=\partial q$ obeys $\iota_Y\omega=dp$, so its charge is $J=p$. Hamilton's equations give $dJ/dt=\dot{p}=-H_q=0$. In variational language, $S=\int(p\,\dot{q}-H)dt$ is invariant under constant q translations; stationarity supplies the equations and conserved momentum. The symmetry, action structure and equations jointly justify conservation. A repeated equal mark or unchanged gap alone does not prove Noether's theorem.

Limits and the remaining debt

Supplied: affine structure, oriented pairing, regularity, model choice and variational hypotheses. Derived under those assumptions: local Hamiltonian form; two exact example flows; the translation charge and its conservation. Not established: the earlier Genesis/L1 derivation, any identity between holonomy and Noether, Born probabilities, Everett branching, Bell correlations, or an ontology of the universe. Those require further constructions and evidence.